TECHNICAL PORTFOLIO

Medical Physics Software and Computational Research

Kai Woodger

Radiotherapy software Treatment planning Computational modelling Verification

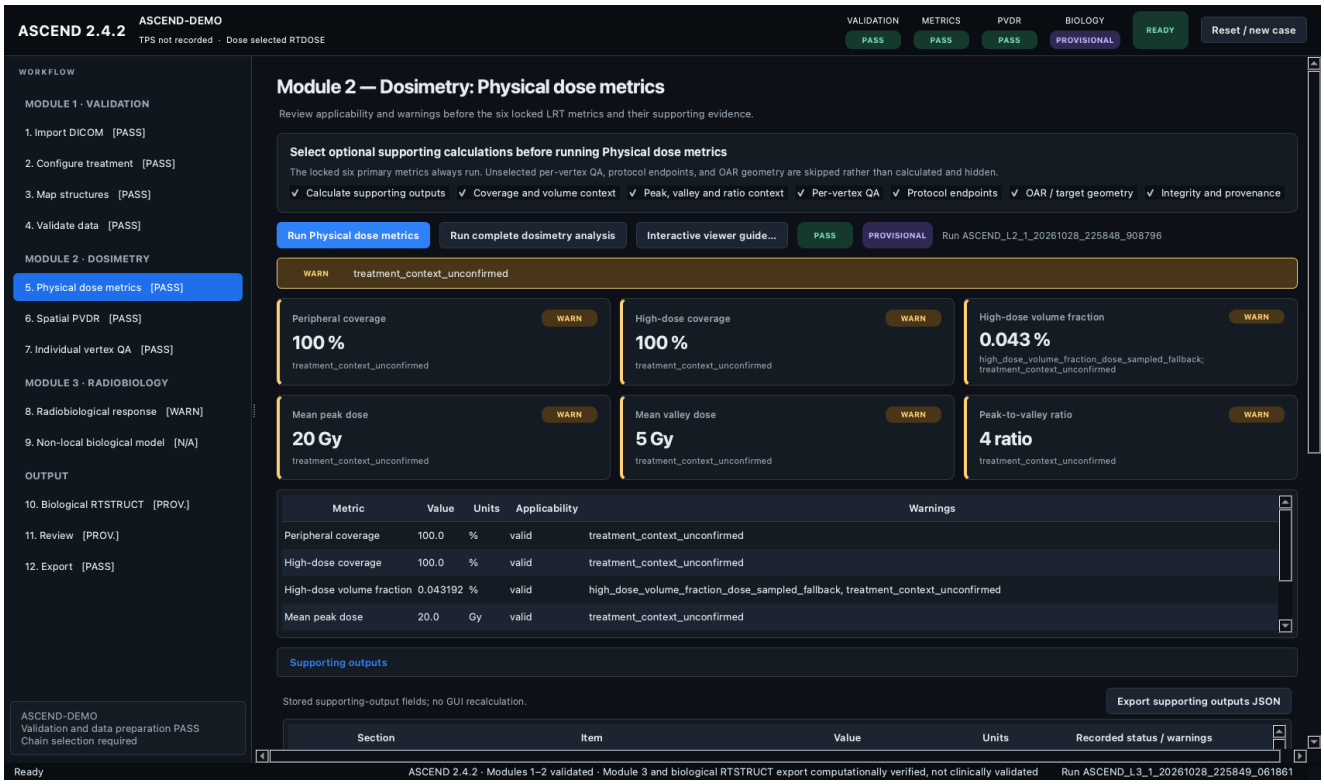


Figure 1 ASCEND 2.4.2 dosimetry workstation using a synthetic demonstration case

November 2026

Technical portfolio with test results and stated clinical limitations

PORTFOLIO SUMMARY

Portfolio Summary

Medical physics projects with supporting test results

I develop research software for radiotherapy plan evaluation, treatment planning, Monte Carlo simulation, biological modelling, and quality assurance. This portfolio explains what each project does, how I tested it, and which clinical questions remain unresolved.

The main project is ASCEND, a desktop application for checking DICOM radiotherapy data and analysing lattice radiotherapy plans. The current version passed 395 tests plus 40 subtests. Earlier release testing also passed on Linux, Windows, and macOS. Numerical reference tests and a comparison with one stored Eclipse case show where the calculations agree and where differences remain.

Area	Evidence in this portfolio
Independent plan analysis	ASCEND 2.4.2 for DICOM checks and lattice radiotherapy dose analysis
Published radiobiology	Peer-reviewed non-local signalling and vascular sink modelling in Physics in Medicine & Biology
Computational modelling	Radiotherapy digital twin using a verified synthetic whole body surface
Treatment planning automation	C sharp ESAPI script for automatic lattice sphere placement
Monte Carlo simulation	TOPAS very high energy electron beamline and beam shape analysis
Quality assurance research	Biological plan audit and uncertainty aware contouring QA prototypes

Interpretation of evidence Passing software tests shows that the implementation behaves as specified. It does not by itself establish clinical accuracy or safety. Each case study states when measurements, treatment planning system comparisons, licensed anatomy data, or clinical outcomes are still required.

CASE STUDY 1

ASCEND Lattice Radiotherapy Workstation

Independent DICOM checks and lattice radiotherapy plan analysis

Repository https://github.com/kaiwooodger/ASCEND

ASCEND is a desktop application for independent analysis of radiotherapy data. Module 1 checks that the CT, structure set, plan, and dose objects belong to the same case and use consistent geometry. Module 2 calculates physical lattice radiotherapy measures, including peak to valley dose ratio and checks for individual high dose vertices. Module 3 contains research models of biological response. These research results are labelled separately because they have not been clinically validated.

Current repository evidence	Value
Release	ASCEND 2.4.2
Current automated tests	395 tests plus 40 subtests passed on macOS ARM64 with Python 3.12
Operating systems	An earlier suite of 229 tests plus 28 subtests passed on six Linux, Windows, and macOS configurations
Repeatability of numerical results	Of 70 reference comparisons, 69 were identical and one differed by 8.88e-16
Comparison with Eclipse	All 19 available dose comparisons passed; five of nine structure volume comparisons passed
Repository size	166 application files, 44 test files, 84 documentation files, and four automated workflows

My contribution

- Defined the complete workflow for DICOM checking, dose analysis, spatial peak to valley dose ratio, biological modelling, and result export.
- Combined the calculation modules, user interface, warnings, reports, and synthetic test cases in one application.
- Managed releases, operating system testing, documentation, and verification records.
- A prior Git audit attributed 99.68 percent of current code lines to Kai Woodger and 52 of 57 non merge commits, while recognising that repository attribution is not a measure of unaided manual typing.

ASCEND VISUAL EVIDENCE

ASCEND Interface and Dose Analysis Method

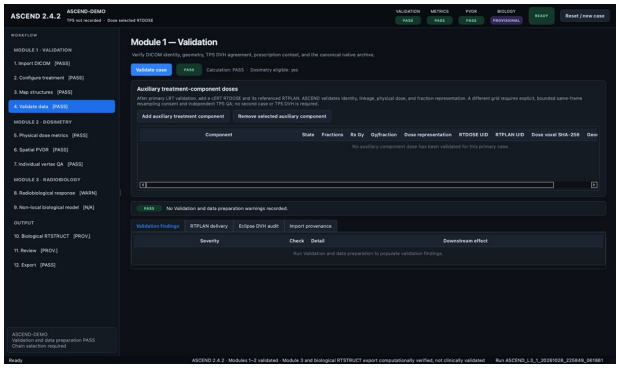


Figure 2 Module 1 validation interface

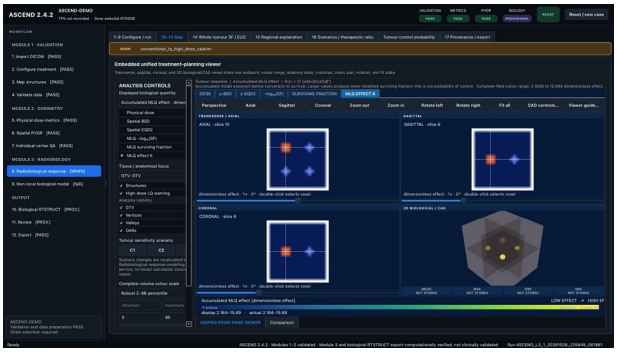


Figure 3 Module 3 research maps

The interface shows which checks have passed, which warnings require review, and which calculations are available. Exported results record the software version, calculation settings, and source geometry so that the analysis can be repeated.

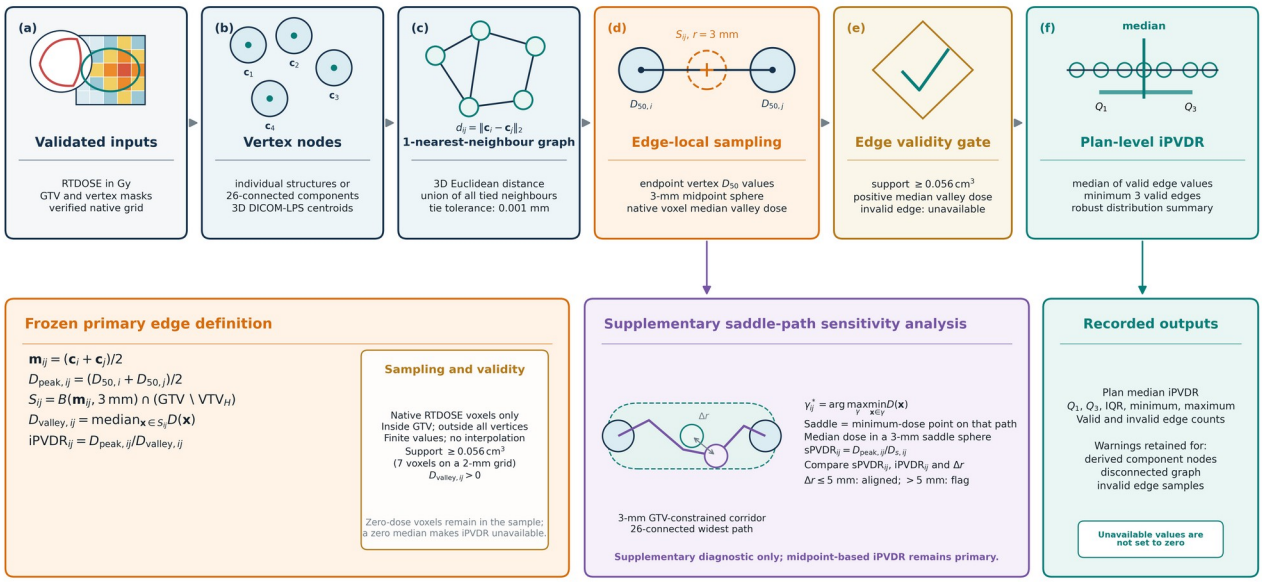


Figure 4 Publication figure for the edge local spatial iPVDR algorithm and supplementary saddle path analysis

ASCEND TESTING

How ASCEND Was Tested and Compared with Eclipse

ASCEND 2.4.2 adds structure specific coverage checks for conventional external beam radiotherapy and keeps conventional and lattice treatment histories separate. The current version passed 395 tests and 40 subtests. An earlier release suite passed 229 tests and 28 subtests on six combinations of Linux, Windows, macOS, Python 3.11, and Python 3.12.

Evidence	Result	Limitation
Current automated tests	395 tests plus 40 subtests passed	Run locally on macOS with Python 3.12
Operating system tests	Six operating system and Python combinations passed	This retained run used the earlier 229 test plus 28 subtest suite
Numerical repeatability	69 of 70 comparisons were identical; largest difference 8.88e-16	Based on 14 synthetic reference measures
Eclipse dose comparison	19 of 19 available comparisons passed	One stored case; eight D2 measures were unavailable
Eclipse structure volumes	Five of nine comparisons passed	Four differences remain; this is not clinical validation

The Eclipse comparison used one stored case with nine structures. Of 28 valid comparisons, 24 passed and four structure volume comparisons failed. All 19 available dose comparisons passed the provisional tolerance of the larger of 0.2 Gy or 2 percent of the Eclipse value. The median absolute dose difference was 0.00643 Gy and the maximum was 0.22245 Gy. This is an initial comparison with an external treatment planning system, not complete clinical validation.

Relevant medical physics skills ASCEND combines DICOM radiotherapy data handling, dose analysis, radiobiological modelling, user interface design, automated testing, documentation, and comparison with Eclipse in one working application.

- DICOM RT identity and geometry handling
- Lattice radiotherapy plan quality and spatial dose analysis
- Desktop application design and guided clinical workflow
- Automated numerical tests and synthetic cases
- Publication figures reports and technical documentation
- Automated release checks on Linux Windows and macOS

CASE STUDY 2

Non-Local Radiobiological Modelling

Peer-reviewed modelling of biological response beyond the high-dose regions

Published article Woodger K. Biology-informed reinterpretation of lattice radiotherapy using non-local bystander signalling and anatomy-aware vascular sink modelling. Physics in Medicine & Biology. 2026;71(16). Published 18 August 2026. doi:10.1088/1361-6560/ae94dd.

DOI record https://doi.org/10.1088/1361-6560/ae94dd

PubMed https://pubmed.ncbi.nlm.nih.gov/42551488/

Reproducibility package https://github.com/kair98-boop/SFRT

I developed a hypothesis-generating computational framework to test whether physical lattice radiotherapy measures retain the same interpretation when biological signals spread beyond the high-dose regions. TOPAS-derived dose distributions were evaluated on a body-aware, smoothed Monte Carlo dose substrate. A reaction-diffusion model propagated ROS-like and cytokine-like fields, while explicit vessel masks produced an anatomy-aware vascular sink field.

The model was calibrated once in a one-dimensional half-field assay and transferred without geometry-specific retuning to stripe, lattice, irregular surrogate, and synthetic head-and-neck geometries. A ten-case algorithmic cohort was evaluated as physical-only, biology-informed without vascular uptake, and biology-informed with anatomical vascular uptake.

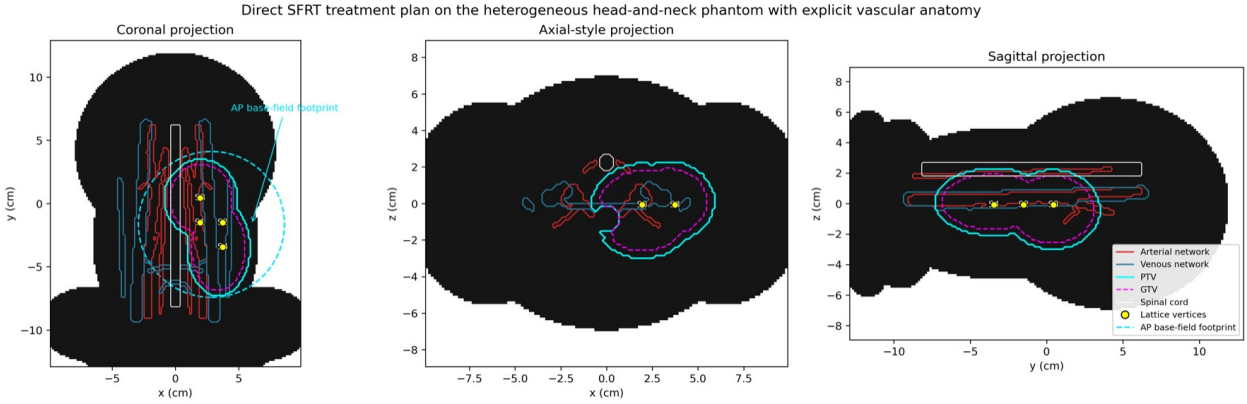


Figure 5 Synthetic head and neck lattice geometry with explicit vascular anatomy

PMB RESULTS

Quantitative Results, Reproducibility, and Limitations

Ten-case comparison of physical and biology-informed interpretations

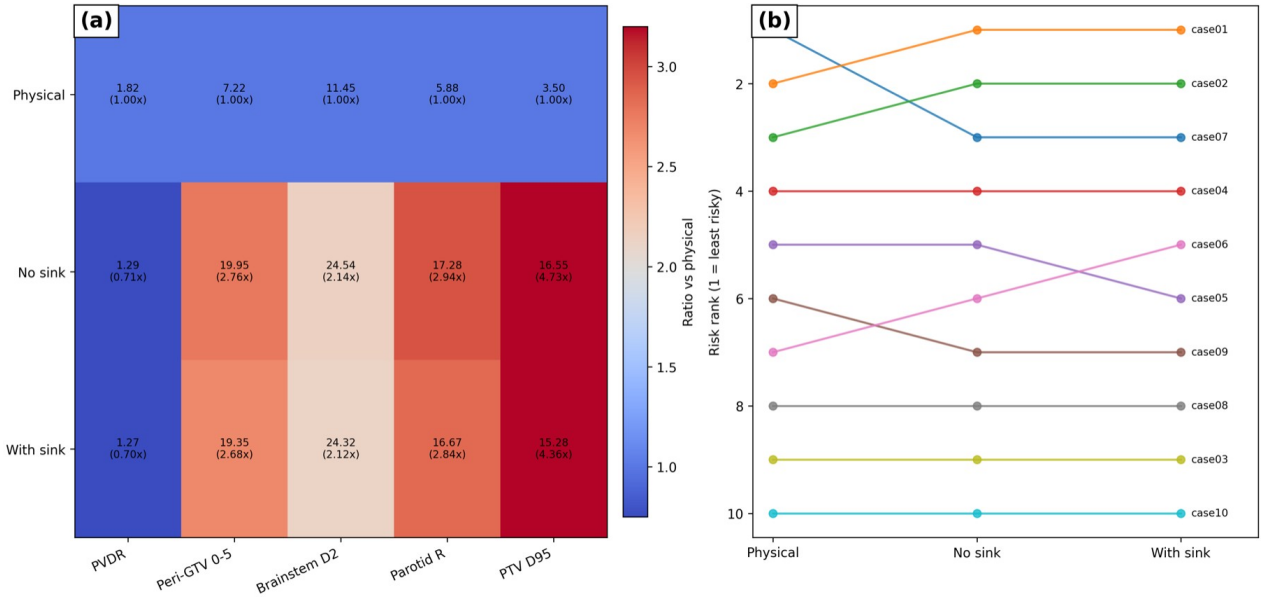


Figure 6 Biological reinterpretation and ranking results across the ten case cohort

Recorded measure	Result
Cohort mean PVDR like ratio	1.817 physical only; 1.292 without vascular uptake; 1.270 with anatomical uptake
Ranking changes versus physical only	5 of 10 without uptake; 6 of 10 with anatomical uptake
Ranking changes between biology modes	2 of 10
Brainstem adjacent biology added burden flags	2 of 10
Sink related shifts above the combined 95 percent uncertainty band	65 of 70
Noise qualified sink driven rank changes	0 of 10

Reproducibility The public package contains Python generators, transport wrappers, biological solvers, endpoint extraction and figure scripts, TOPAS templates, locked-parameter YAML snapshots, manifests, and a bounded paper-regeneration check. Oversized raw transport intermediates are excluded, while the scripts and curated outputs required to inspect the workflow and regenerate the publication-facing evidence are retained.

Clinical limitation The study is computational and hypothesis-generating. It does not establish clinical benefit. Patient-derived RTDOSE and RTSTRUCT data, realistic vascular segmentation, and independent experimental validation are required before clinical use.

CASE STUDY 3

Radiotherapy Digital Twin Research Prototype

A repeatable simulation of anatomy motion circulation radiation and biology

This project tests how several radiotherapy simulation components can work together. It combines a synthetic whole body model with DICOM patient coordinates, deformation, blood and lymph transport, radiation energy deposition, oxygen response, and biological calculations. The software records calculation settings and can restart from saved checkpoints.



Figure 7 Watertight synthetic whole body mesh

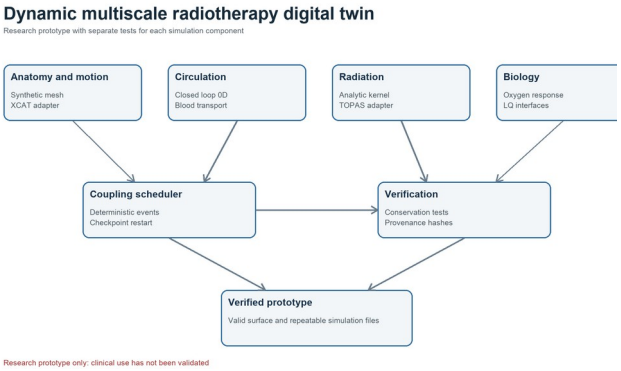


Figure 8 Implemented system architecture

Engineering release evidence	Result
Mesh	39,664 triangles
Surface quality	No holes nonmanifold edges or degenerate elements
Automated tests	8 passed in the immutable v0.1.0 release
Repeatability	A clean rerun reproduced the same simulation arrays summaries and surface file
Clinical status	Research prototype only and not validated for clinical use

DIGITAL TWIN TESTING

External Models and Remaining Validation Work

The project uses separate reference models for different parts of the simulation. HEDOS is used to test blood transport and blood dose calculations. XCAT can test anatomy and motion when the licensed software and reference data are available. TOPAS is used for radiation transport. Agreement in one component does not validate the other components or the complete model.

Reference	Result	What remains
HEDOS	Twenty random seeds and 100,000 particles per seed reproduced field and DVH based calculations	This does not validate human physiology
XCAT	The data interface passed 24 of 24 package intake checks	Full execution requires a provider licence and Windows x86 64
TOPAS	A fixed test case confirmed that the interface can run	Independent measurement and local commissioning are still required
Combined simulation	Dose conservation restart and grid refinement checks passed	Only synthetic component interfaces have been tested

My contribution

- Converted the radiotherapy digital twin concept into separate testable modules with defined inputs outputs and acceptance criteria.
- Built repeatable calculations for geometry physiology transport radiation and biological interfaces.
- Recorded the licensing platform and clinical evidence requirements that still limit the project.
- Advanced the later repository to 65 passing tests during authorised XCAT intake while retaining the clinical block.

CASE STUDY 4

Automatic Lattice Sphere Placement in Eclipse

C sharp treatment planning automation with independent geometry checks

Repository https://github.com/kaiwooodger/ESAPI-Script

The C sharp script automatically places a three dimensional lattice of spheres in an Eclipse structure set. It creates the individual sphere structures and their union while leaving the original target unchanged. Before saving the result, it checks the contracted target boundary, image limits, sphere spacing, containment, names, and cleanup of incomplete outputs.

Method reference The sphere placement geometry follows Section 2.3 of Gaudreault M, Chang D, Kron T, Siva S, Chander S, Hardcastle N and Yeo A, Development of an automated treatment planning approach for lattice radiation therapy, Medical Physics. 2024;51(1):682-693. doi:10.1002/mp.16761.

Open method paper https://doi.org/10.1002/mp.16761

The retry order, superior and inferior end fill search, and safety checks are implementation choices. This portfolio does not claim to reproduce the authors' source code.

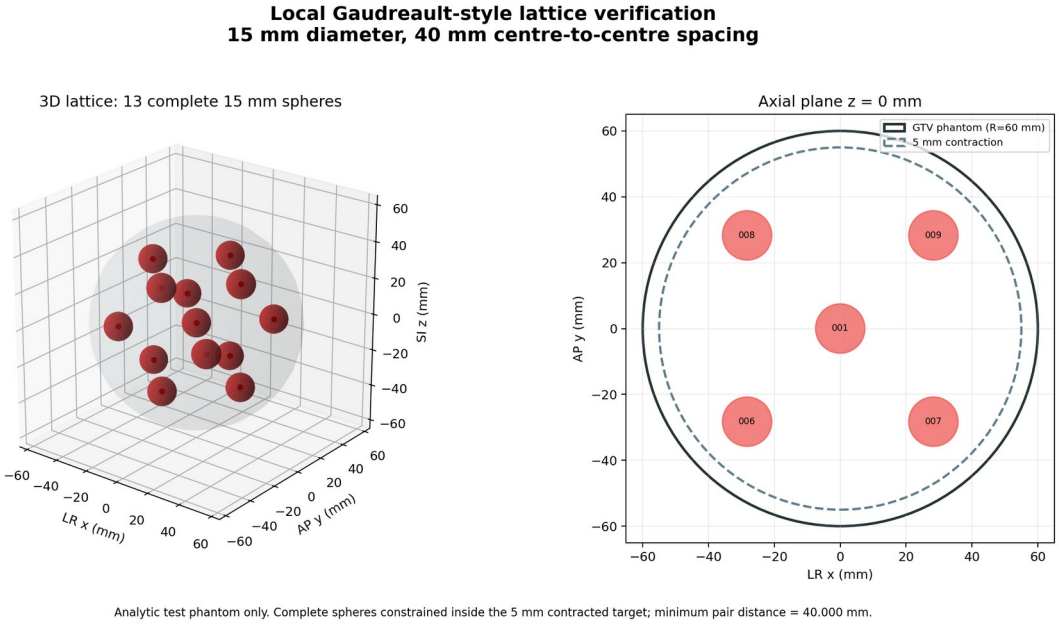


Figure 9 Local phantom verification for 15 mm spheres with 40 mm centre to centre spacing

- The verification program used the same geometry code as the Eclipse script.
- A worked phantom produced 13 complete spheres with a 40.000000 mm minimum pair distance.
- The original GTV remains unchanged and existing output structure names stop the script before modification.
- Execution inside Eclipse and compilation against the installed Varian libraries must still be verified at each installation.

ESAPI TESTING

Geometry Tests Across Target Sizes and Sphere Spacings

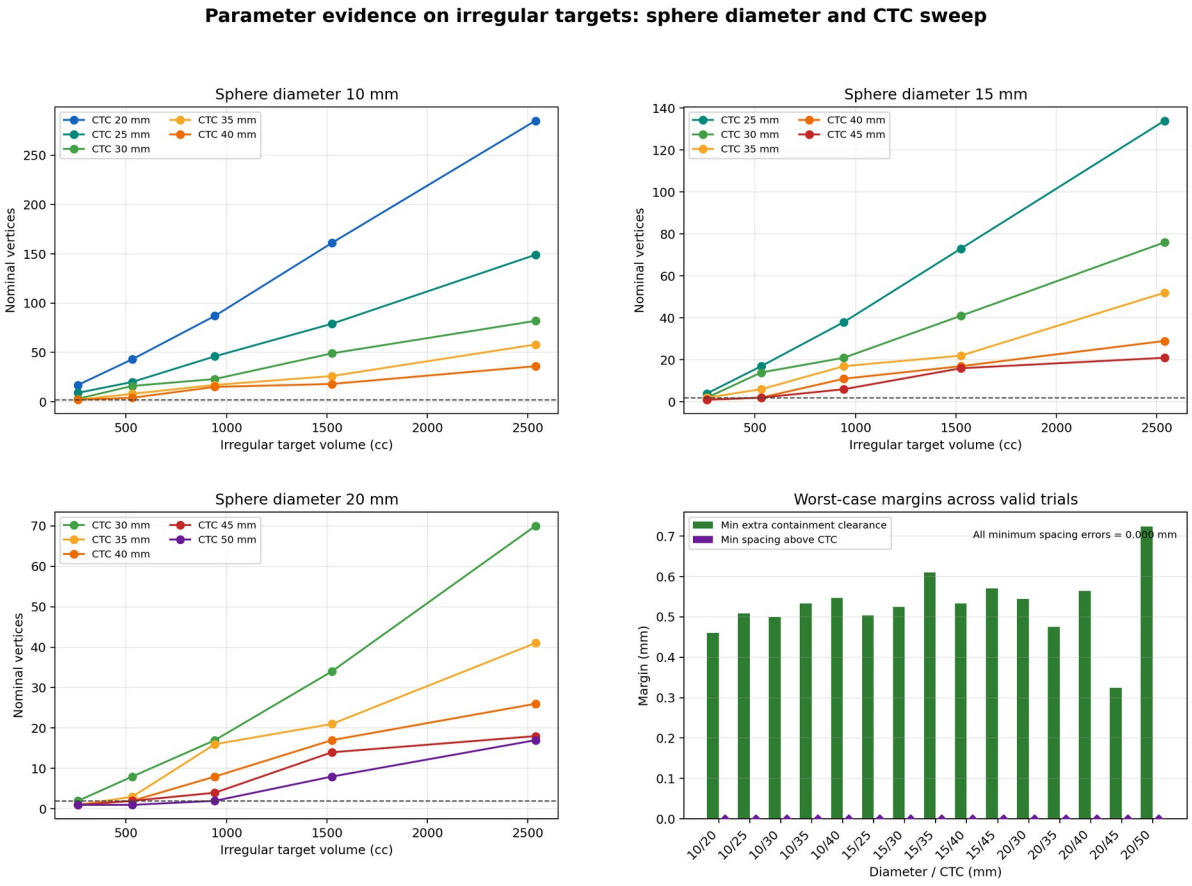


Figure 10 Sphere diameter and centre to centre spacing evidence across irregular target sizes

Verification measure	Recorded result
Total trials	375
Valid lattices	336
Correct no lattice outcomes	39
Geometric failures	0
Worst spacing error	0.000000 mm
Minimum containment clearance	0.324 mm
Maximum tested lattice	292 vertices
Core regression suite	19 passed

ADDITIONAL PROJECTS

Biological Plan Audit Contouring QA and Electron Beam Simulation

Research prototypes and recorded test results



Figure 11 BARAT verification evidence

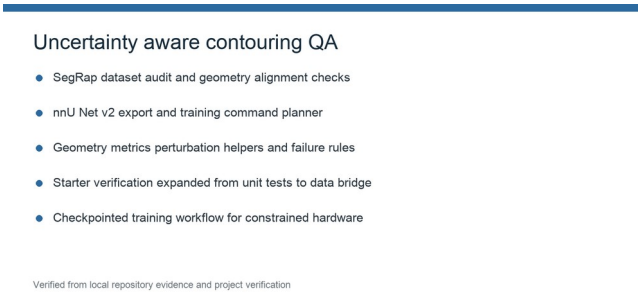


Figure 12 Contouring QA verification evidence

BARAT Biological Audit and Risk Analysis Tool

BARAT is a research prototype for biological review of radiotherapy plans. Its command line and dashboard interfaces show whether the input data model and calibration are ready before plan comparison or report export. The recorded test runs reached 63 to 64 passing tests. The software is not intended for clinical use.

Uncertainty Aware Contouring QA

This project assesses automatic contours using stability and anatomical plausibility as well as overlap measures such as Dice similarity. It includes geometric measures, controlled contour perturbations, failure criteria, SegRap data checks, nnU Net v2 data export, and a training workflow that can resume from saved checkpoints.

SIMULATION RECORD

TOPAS Very High Energy Electron Beamline Modelling

This project uses TOPAS and Python to model a very high energy electron beamline with four quadrupole magnets at 100 200 and 250 MeV. The analysis varies the fourth quadrupole strength and the source model, then measures depth dose and the Gaussian beam width in the two transverse directions.

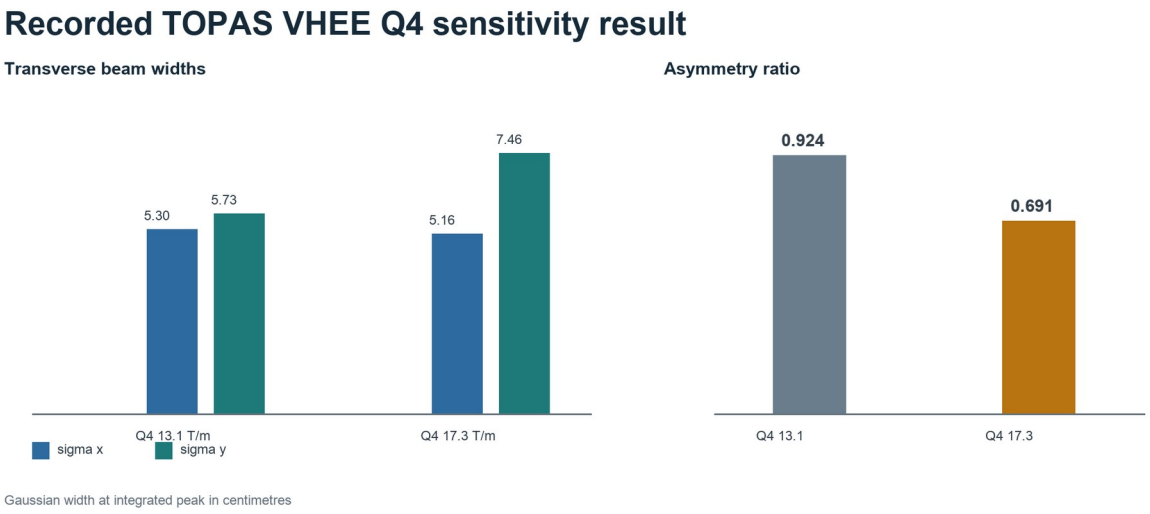


Figure 13 Recorded 250 MeV Q4 sensitivity result using the 4.0 mm source sigma baseline

Increasing the fourth quadrupole strength from 13.1 to 17.3 T per m increased the difference between the horizontal and vertical beam widths. At the integrated peak the width ratio changed from 0.924 to 0.691. A percentage depth dose curve alone would not show this transverse asymmetry.

Q4 T per m	Integrated peak cm	Sigma x cm	Sigma y cm	Ratio
13.1	23.9109	5.2958	5.7307	0.924
17.3	16.7822	5.1558	7.4577	0.691

PROFESSIONAL SUMMARY

Medical Physics and Software Skills

Examples supported by the projects in this portfolio

Capability	Demonstrated evidence
Independent plan analysis	DICOM checks lattice radiotherapy dose measures spatial peak to valley dose ratio radiobiology and reporting
Scientific computing	Python NumPy SciPy PySide6 PyVista pydicom R and repeatable numerical calculations
Treatment planning	Eclipse ESAPI C sharp automation DICOM structure handling and synthetic phantom tests
Computational modelling	Anatomy motion circulation blood and lymph transport radiation oxygen and biological response
Monte Carlo simulation	TOPAS beamline models parameter studies and dose grid analysis
Machine learning QA	SegRap data checks nnU Net v2 export uncertainty measures failure criteria and resumable training
Software testing	Modular design automated tests release checks calculation records and documentation
Research communication	Publication figures thesis material validation reports abstracts and technical documentation

Resume ready project bullets

- Developed ASCEND 2.4.2 for independent DICOM checks and lattice radiotherapy plan analysis. The current version passed 395 tests plus 40 subtests, and all 19 available dose comparisons passed the provisional Eclipse agreement rule.
- Published a Physics in Medicine & Biology study of non-local signalling and anatomy-aware vascular uptake in lattice radiotherapy. Across ten controlled cases, the cohort mean PVDR-like ratio changed from 1.817 under physical interpretation to 1.292 without uptake and 1.270 with anatomical uptake.
- Designed a radiotherapy digital twin research prototype that combines synthetic anatomy motion circulation radiation transport oxygen response and biological calculations in a repeatable simulation.
- Implemented and tested a C sharp ESAPI script for automatic lattice sphere placement across 375 irregular target trials with no geometry failures among 336 valid lattices.
- Built a TOPAS very high energy electron beamline model and quantified how energy and quadrupole settings changed transverse beam asymmetry and depth dose.
- Created research prototypes for biological plan audit and contouring QA with data readiness checks uncertainty measures nnU Net integration and explicit clinical limitations.